Cardiac Physiology

Cardiovascular System - Lecture 3

Department of Physiology

Second Year Medicine

Learning Objectives

- Describe the conduction system of the heart
- Explain the phases of the cardiac cycle
- Calculate cardiac output and explain how it is regulated

Functional Anatomy of the Heart

- The heart has four chambers: two atria that receive blood and two ventricles that pump it out.
- The right ventricle pumps deoxygenated blood into the pulmonary trunk; the left ventricle pumps oxygenated blood into the aorta.
- The wall of the left ventricle is about three times thicker than that of the right ventricle because it pumps against a higher pressure.
- The tricuspid valve separates the right atrium from the right ventricle; the mitral (bicuspid) valve separates the left atrium from the left ventricle.
- The pulmonary and aortic valves are semilunar valves that prevent backflow from the arteries into the ventricles.

Cardiac Muscle Cells

- Cardiac muscle cells are striated, branched and usually have one central nucleus.
- Intercalated discs join neighbouring cells; their gap junctions let action potentials spread from cell to cell.
- Desmosomes in the intercalated discs hold the cells together during contraction.
- Cardiac muscle has a long refractory period, so it cannot be tetanized.

The Conduction System

- The sinoatrial (SA) node in the wall of the right atrium is the normal pacemaker of the heart.
- The impulse spreads through the atrial muscle to the atrioventricular (AV) node.
- The AV node delays the impulse by about 0.1 second, so the atria finish contracting before the ventricles contract.
- From the AV node the impulse passes to the bundle of His, the right and left bundle branches, and the Purkinje fibres.
- Purkinje fibres conduct the impulse fastest and spread it through the ventricular muscle from the apex upward.

Pacemaker Potential

- SA node cells have no stable resting potential; they depolarize slowly toward threshold.
- The slow depolarization is caused by the funny current (If), an inward sodium current through HCN channels.
- At threshold, calcium enters through L-type calcium channels and produces the upstroke of the action potential.
- Repolarization is caused by the opening of potassium channels.

Ventricular Action Potential

- Phase 0: rapid depolarization as fast voltage-gated sodium channels open.
- Phase 1: brief early repolarization as the sodium channels inactivate and transient potassium channels open.
- Phase 2: the plateau, in which calcium entry through L-type channels balances potassium exit.
- Phase 3: repolarization as the calcium channels close and potassium efflux increases.
- Phase 4: the resting membrane potential of about -90 mV.

The Electrocardiogram (ECG)

- The P wave represents atrial depolarization.
- The QRS complex represents ventricular depolarization; atrial repolarization is hidden within it.
- The T wave represents ventricular repolarization.
- The PR interval, normally 0.12 to 0.20 second, reflects conduction from the atria through the AV node.
- A QRS complex longer than 0.12 second suggests delayed conduction within the ventricles.

The Cardiac Cycle

- Atrial systole pushes the last 20 to 30% of ventricular filling into the ventricles.
- Isovolumetric contraction begins when the atrioventricular valves close; all four valves are closed and the ventricular volume does not change.
- Ventricular ejection begins when ventricular pressure exceeds arterial pressure and the semilunar valves open.
- Isovolumetric relaxation begins when the semilunar valves close; ventricular pressure falls while all valves are closed.
- Ventricular filling begins when ventricular pressure falls below atrial pressure and the atrioventricular valves open.

Heart Sounds

- The first heart sound (S1) is caused by closure of the atrioventricular valves at the start of ventricular systole.
- The second heart sound (S2) is caused by closure of the semilunar valves at the start of ventricular diastole.
- A third heart sound (S3) may be heard during rapid ventricular filling and is normal in children.
- Murmurs are abnormal sounds caused by turbulent blood flow, for example through a narrowed (stenotic) valve.

Cardiac Output

- Cardiac output is the volume of blood each ventricle pumps per minute: cardiac output = heart rate x stroke volume.
- At rest, a heart rate of 70 beats per minute and a stroke volume of 70 mL give a cardiac output of about 4.9 litres per minute.
- Stroke volume = end-diastolic volume - end-systolic volume, for example 120 mL - 50 mL = 70 mL.
- Ejection fraction = stroke volume / end-diastolic volume; it is normally about 55 to 70%.

Regulation of Stroke Volume

- Preload is the degree of stretch of the ventricular muscle before contraction; it rises when venous return rises.
- Frank-Starling law: the greater the stretch of the ventricle in diastole, the stronger its contraction and the larger the stroke volume.
- Contractility is the force of contraction at a given preload; sympathetic stimulation increases it.
- Afterload is the pressure the ventricle must overcome to eject blood; a high arterial pressure increases afterload and reduces stroke volume.

Regulation of Heart Rate

- Sympathetic nerves release noradrenaline, which acts on beta-1 receptors to increase heart rate and contractility.
- Parasympathetic fibres of the vagus nerve release acetylcholine, which acts on M2 receptors to slow the heart rate.
- At rest the vagus nerve dominates, so the resting heart rate is lower than the intrinsic SA node rate of about 100 beats per minute.
- Adrenaline from the adrenal medulla, a high body temperature and thyroid hormone also increase heart rate.

Check Your Understanding

1. A patient's heart rate is 80 beats per minute and the stroke volume is 60 mL. What is the cardiac output?
2. Which valves close at the start of isovolumetric contraction?

Thank You

Questions?